Dengue Fever — Clinical Management Guide

1. Overview and Epidemiology

Dengue fever is a mosquito-borne viral infection caused by four closely related serotypes (DENV 1-4) of the genus Flavivirus, transmitted primarily by Aedes aegypti mosquitoes. With over 400 million infections annually, dengue is one of the world's most prevalent vector-borne diseases.

India experiences some of the world's highest dengue burdens, with major urban epidemics in Delhi, Mumbai, Chennai, and Hyderabad. Cases peak after the monsoon season (August-November) when mosquito breeding is at its highest.

Climate change, rapid urbanization, inadequate water storage practices, and poor sanitation contribute to the expansion of Aedes aegypti habitats and year-round dengue transmission in tropical cities.

2. Virology and Transmission

There are four dengue serotypes (DENV-1, DENV-2, DENV-3, DENV-4). Infection with one serotype provides lifelong immunity to that serotype but only temporary cross-immunity to the others. A second infection with a different serotype carries a significantly higher risk of severe dengue.

Aedes aegypti is the primary vector — a day-biting urban mosquito that breeds in clean, standing water in containers (flower pots, coolers, discarded tyres, water storage tanks). The Aedes albopictus mosquito also transmits dengue but is a less efficient vector.

The extrinsic incubation period in the mosquito is 8-12 days at 25-28°C. The intrinsic incubation period in humans is 4-10 days after a mosquito bite. Dengue is NOT transmitted directly from person to person except through blood transfusions and organ transplants.

3. Clinical Phases

Dengue illness progresses through three overlapping phases that healthcare providers must recognize:

Phase 1 — Febrile Phase (Days 1-3): Abrupt high fever (39-40°C), severe headache, retro-orbital pain (pain behind eyes worsened by eye movement), myalgia and arthralgia (breakbone fever), flushed face, early petechiae. The fever may be biphasic.

Phase 2 — Critical Phase (Days 4-6): As the fever breaks, plasma leakage from blood vessels due to increased capillary permeability can cause: hemoconcentration (rising hematocrit ≥20%), pleural effusion, ascites, and hypotension leading to dengue shock. Platelet count falls sharply. Warning signs must be watched for.

Phase 3 — Recovery Phase (Days 7-10): Reabsorption of leaked fluids. Risk of fluid overload if IV fluids given excessively. Platelet count begins to rise. Bradycardia may occur. Characteristic convalescent rash — islands of white in a sea of red.

4. Warning Signs of Severe Dengue

WHO dengue warning signs requiring urgent hospitalization: abdominal pain or tenderness, persistent vomiting, clinical fluid accumulation (ascites, pleural effusion), mucosal bleeding (gum bleed, epistaxis), lethargy or restlessness, liver enlargement >2 cm, and rising hematocrit with rapid decrease in platelet count.

Severe dengue (formerly dengue hemorrhagic fever/dengue shock syndrome): severe plasma leakage → dengue shock syndrome (DSS), severe bleeding, and severe organ impairment (liver — AST/ALT >1000, CNS involvement, myocarditis).

Signs of dengue shock: cold, clammy skin; rapid weak pulse; narrowed pulse pressure (<20 mmHg); hypotension; decreased urine output; anxiety. This is a medical emergency with up to 10-20% mortality if untreated, but <1% with prompt fluid replacement.

5. Laboratory Findings

Test	Typical Findings in Dengue
Complete Blood Count	Leukopenia, rising hematocrit (critical phase), thrombocytopenia
Dengue NS1 Antigen	Positive Days 1-5 (early diagnosis)
Dengue IgM Antibody	Positive from Day 5-7 onwards (primary infection)

Dengue IgG Antibody	Elevated in secondary infection; persists for years
Liver Function Tests	Elevated AST/ALT, especially in severe dengue
Coagulation (PT/APTT)	Prolonged in severe dengue with bleeding
Chest X-ray/Ultrasound	Pleural effusion, ascites, gallbladder oedema

6. Treatment and Fluid Management

There is no specific antiviral treatment for dengue. Management is supportive and centres on fluid management, monitoring, and early identification of the critical phase.

Outpatient management (non-severe dengue without warning signs): paracetamol 500-1000 mg every 6 hours for fever and pain. Ensure oral fluid intake of at least 2-3 litres/day (water, ORS, coconut water, fruit juices). Daily CBC monitoring. Return immediately if warning signs appear.

STRICTLY AVOID: ibuprofen, aspirin, naproxen, diclofenac — NSAIDs increase bleeding risk, damage gastric mucosa, and can precipitate Reye's syndrome in children. Avoid intramuscular injections.

Inpatient fluid management: IV crystalloids (Normal Saline or Ringer's Lactate) guided by clinical signs, urine output (target 0.5-1 ml/kg/hr), hematocrit, and vital signs. Initial rate: 5-7 ml/kg/hr, titrate down as patient improves. Overhydration causes pulmonary oedema.

Platelet transfusion: platelet count <10,000/mm^3 without bleeding, or <20,000-50,000 with active bleeding or planned invasive procedure. Do NOT transfuse platelets prophylactically at higher counts — no survival benefit and risk of fluid overload.

7. Emergency Actions for Dengue Shock

Dengue shock is a medical emergency. Move to hospital immediately. Secure IV access. Give rapid fluid bolus — NS or RL 10-20 ml/kg over 15-30 minutes. Reassess after each bolus.

Monitor: vital signs every 15-30 min during resuscitation, hematocrit every 4 hours, urine output hourly. Falling hematocrit + hemodynamic instability = internal bleeding — consider blood transfusion.

Oxygen supplementation: maintain SpO2 >95%. Avoid excess fluid in recovery phase — rising hematocrit resolving, abundant urine output, bradycardia = fluid reabsorption phase.

Reduce IV fluids to avoid pulmonary oedema.

8. Prevention and Vector Control

Personal protection: use DEET (N,N-diethyl-meta-toluamide) based repellents on exposed skin. Wear light-coloured, full-sleeve clothing. Use mosquito nets, especially for young children and the elderly. Window screens provide passive protection.

Eliminating breeding sites: the most effective community intervention. Empty and scrub all water containers weekly (Aedes can complete breeding in as little as 7 days in clean stagnant water). Cover water storage tanks. Remove discarded tyres, coconut shells, and flowerpot trays.

Community and government measures: outdoor fogging with insecticides (provides short-term relief). Biological control (Bacillus thuringiensis israelensis in standing water). Sterile Insect Technique (SIT) and Wolbachia-infected Aedes mosquitoes are emerging biocontrol methods being trialled.

The dengue vaccine Dengvaxia (CYD-TDV, Sanofi) is approved for use in dengue-seropositive individuals 9-45 years in endemic areas. It is NOT recommended for dengue-naive individuals (can cause severe dengue on first natural infection). A pre-vaccination screening test is mandatory.

9. Special Populations

Children under 15: dengue shock syndrome is more common in children. Fluid management must account for weight. Younger children may not express symptoms clearly — parental observation is critical. School outbreaks are common.

Pregnancy: dengue in pregnancy associated with premature birth, maternal haemorrhage, low birth weight, and vertical transmission near delivery. Platelet transfusion may be required before delivery. Paracetamol is the only safe analgesic.

Elderly patients: more susceptible to severe dengue and complications (cardiac involvement, haemorrhage). Fluid balance is challenging — prone to overload. High mortality if dengue occurs with pre-existing heart failure or renal impairment.

Second dengue infection: significantly higher risk of severe dengue due to antibody-dependent enhancement (ADE) — pre-existing antibodies from the first infection facilitate viral uptake by macrophages, increasing viral load. All patients with possible second infection should be monitored more closely.

10. Differential Diagnosis and India-Specific Notes

Dengue must be distinguished from: malaria (common in same season, geographic overlap in India), leptospirosis (post-flood, rat exposure, conjunctival suffusion, renal involvement), scrub typhus (eschar, lymphadenopathy, rural exposure), typhoid (relative bradycardia, rose spots, abdominal symptoms), chikungunya (prolonged arthralgia, rash, rare shock), and viral hepatitis.

In India, co-infections of dengue with malaria and leptospirosis are well-documented and associated with higher mortality. A dengue diagnosis should not preclude testing for other concurrent infections.

India's National Dengue Day (May 16) focuses on awareness. The National Vector Borne Disease Control Programme (NVBDCP) coordinates surveillance and control. Timely reporting of dengue cases is essential for outbreak detection.

Private labs vary widely in dengue test quality. A single negative rapid NS1 test on day 1-2 does not rule out dengue — repeat testing at Day 4-5 or send IgM ELISA. RT-PCR is the gold standard for early diagnosis in reference labs.